BIOSZEN User Manual (English)

This manual explains how to run BIOSZEN workflows from data import to reproducible export.

1. Prerequisites

- R >= 4.1 installed.
- BIOSZEN app launched from `App.R` or `BIOSZEN::run_app()`.
- Input files for **Load Data** can be `Excel` (`.xlsx`, `.xls`) or `CSV` (`.csv`).
- Curve files for **Load Curves** can be `Excel` (`.xlsx`, `.xls`) or `CSV` (`.csv`).

Reference templates are available in the app under **Reference input files (download)**:

- `Ejemplo_platemap_parametros.xlsx`
- `Ejemplo_curvas.xlsx`
- `Ejemplo_parametros_agrupados.xlsx`
- `Ejemplo_input_summary_mean_sd.xlsx`

2. Input Modes

2.1 Recommended mode: Platemap + Curves

Use this mode for full functionality and strongest statistical support.

1. Upload a platemap file in **Load Data**.
2. Upload a curves file in **Load Curves** (`.xlsx`, `.xls`, or `.csv`).

Platemap requirements:

- Sheet `Datos`: `Well`, `Strain`, `Media`, `BiologicalReplicate`, `TechnicalReplicate`, `Orden`, plus one or more parameter columns.
- Sheet `PlotSettings`: `Parameter`, `Y_Max`, `Interval`, `Y_Title`.

Curves requirements:

- In `Excel` (`.xlsx`, `.xls`): `Sheet1`: first column `Time`, additional columns by well (`A1`, `A2`, etc.). `Sheet2`: `X_Max`, `Interval_X`, `Y_Max`, `Interval_Y`, `X_Title`, `Y_Title`.
- In `CSV` (`.csv`): first column `Time`, additional columns by well (`A1`, `A2`, etc.). This is a single-table format (no extra sheets). Axis settings are auto-generated: `X_Max` and `Y_Max` use observed maxima, `Interval_X` and `Interval_Y` use `max/4`. `X_Title` and `Y_Title` are blank by default.

2.2 Grouped parameters mode (parameters only)

- Upload grouped-parameter workbook in **Load Data**.
- Use when you only need parameter charts and stats.
- Not valid as a curves input.

2.3 Single-file summary mode (Mean/SD/N)

- Upload summary workbook in **Load Data**.

- Parameters and curves can be detected from dedicated summary sheets.
- Useful when raw replicate rows are unavailable.

2.4 CSV mode (high-volume datasets)

- **Load Data** supports `.csv` files.
- Recommended for larger datasets because it is a lighter format.
- BIOSZEN auto-detects common delimiters (`,`, `;`, tab, or `|`).
- If CSV structure is not already platemap-ready, BIOSZEN attempts conversion to a compatible working profile.
- **Load Curves** also supports `.csv` for well-based trajectories (`Time` + well columns).
- Metadata files remain `.xlsx`.

3. Standard Workflow

1. Load primary data file.
2. Optionally load/merge curves.
3. Optionally load metadata file.
4. Choose scope (`By Strain` or `Combined`).
5. Choose plot type.
6. Apply filters and replicate selections.
7. Optionally normalize by control.
8. Run statistical analysis.
9. Add significance annotations.
10. Export plots, tables, metadata, and bundle.

4. Plot Types and How to Use Them

Boxplot

- Best for raw replicate distributions.
- Control spread with jitter, box width, and point size.
- Add significance bars or labels after stats.

Barplot

- Best for summarized comparison by group.
- Supports mean with error bars and optional raw points (raw mode).
- Supports manual and automatic significance annotations.

Violin

- Best for distribution shape with replicate overlays.
- Supports same annotation workflow as boxplot/barplot.

Stacked

- Use parameter selector and parameter order controls.
- Configure deviation bars and parameter color behavior.
- Supports label or bracket significance mode.

Correlation

- Select X and Y parameters.
- Choose Pearson, Spearman, or Kendall.
- Optional overlays: regression line, r, p, R2, equation.
- Advanced panel supports anchor-based one-vs-all screening and Excel export.

Heatmap

- Select parameter subset.
- Scale mode: none, by row, or by column.
- Optional row/column clustering and dendrograms.
- Optional in-cell value labels.
- Cluster exports available when row clustering is enabled.

Correlation Matrix

- Multi-select parameters.
- Choose correlation method.
- Apply multiple-testing correction (Holm, FDR, Bonferroni, none).
- Optionally display only significant labels.

Curves

- Configure axis limits, labels, and line width.
- Choose line geometry and confidence interval style.
- Optional replicate trajectory display in raw mode.

Composition Panel (multi-plot layouts)

Recommended flow:

1. From each individual chart, click **Add to panel**.
2. Open the **Composition Panel** tab.
3. In **1) Plot Selection**, choose plots to combine and adjust order (move up/down, move to top/bottom, or remove).
4. In **2) Layout & Canvas**, define rows/columns, optional custom grid, column/row sizes, and final output size in px.
5. In **3) Visual Style**, configure legend mode/side, fonts, global text sizes, and palette.
6. Optional: add rich text boxes and per-plot overrides.
7. Preview and export to `PNG`, `PPTX`, or `PDF`.

Composition metadata:

- **Download metadata** stores panel layout and styling.
- **Upload composition metadata (.xlsx)** restores a composition in a new session.

5. Normalization

Enable **Normalize by control** and choose a control medium.

- BIOSZEN computes normalized columns with `_Norm` suffix.

- Correlation supports axis-specific normalization (`both`, `X only`, `Y only`).
- If strict control pairing is not possible, BIOSZEN applies fallback logic.

6. Statistics

Normality tests

- Shapiro-Wilk
- Kolmogorov-Smirnov
- Anderson-Darling

Significance tests

- ANOVA
- Kruskal-Wallis
- t-test
- Wilcoxon

Comparison modes:

- All vs all
- Control vs all
- Pair

Multiple-testing correction:

- Holm
- FDR
- Bonferroni
- None

Summary mode notes:

- Normality may be unavailable (`NA`).
- Some non-parametric paths requiring raw observations are disabled.

7. Significance Annotation Workflow

Manual

1. Select Group 1 and Group 2.
2. Enter annotation label (for example `*`, `**`, `***`, `ns`).
3. Add, reorder, edit, or remove annotations.

Automatic

1. Run significance tests.
2. Open auto-annotation options.
3. Choose inclusion (`significant only` or `all`).
4. Choose label mode (`stars` or `p-value`).
5. Replace or append existing annotations.

8. QC and Replicate Management

Use QC panels to review:

- Missing values.
- Outliers by group.
- Sample size and replicate coverage.

Biological replicates

- Manual include/exclude controls.
- Automatic IQR-based filtering.
- Keep-N reproducibility-oriented selection.

Keep-N automatic selection:

- Ranks replicates by reproducibility (distance to the group median across parameters).
- Keeps the closest replicates to that center (lowest score).
- Combine it with IQR outlier filtering when you want stricter replicate selection.

Technical replicates (inside each biological replicate)

- Dedicated **Technical replicate QC** tab (shown when valid technical replicates exist).
- Group and biological-replicate selectors to view and manually deselect technical replicates.
- Global buttons to select all or deselect all technical replicates.
- Automatic technical outlier detection with IQR.
- Technical Keep-N to keep the N most reproducible technical replicates per subgroup.

9. Metadata, Versioning, and Reproducibility

Metadata workflow:

- Export current UI state with **Download metadata**.
- Re-import metadata to restore compatible settings.

Versioning and bundle workflow:

- Save plot versions in-session.
- Build reproducibility ZIP with plot assets and metadata.
- Reopen work with consistent configuration.

10. Downloads

Main outputs include:

- Plot image (`PNG`, `PDF` depending on plot type).
- Data export.
- Metadata export.
- Statistics export.
- Bundle ZIP.
- Advanced correlation table export.
- Merged platemap/curves exports (when merge tools are used).

11. Growth Module

The growth tab extracts growth parameters from uploaded files:

- `uMax`
- `max_percap_time`
- `doub_time`
- `lag_time`
- `ODmax`
- `max_time`
- `AUC`

Typical use:

1. Upload one or more growth files.
2. Set max time and interval.
3. Run extraction.
4. Download ZIP output.
5. Import extracted outputs into plotting workflow when needed.

12. Troubleshooting

- **Upload error**: verify required sheets and column names.
- **No plot generated**: confirm selected strain/parameter is present after filters.
- **Normalization unavailable**: verify control medium exists in current scope.
- **Stats disabled**: check whether your input mode supports that test path.
- **Curve merge issues**: ensure platemap merge is loaded first for well remapping.
- **CSV not recognized**: validate required headers and a consistent delimiter (`data .csv`: `Strain`/`Media`; `curves .csv`: `Time` + well columns).
- **Slow performance**: reduce selected parameters, disable heavy overlays, or filter groups.

13. Support

For support or bug reports: `bioszenf@gmail.com`